

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459475.8881 +/- 0.00046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.048
Transit depth (Rp/Rs)^2	2.75 +/- 1.6 [%]
Orbital Inclination (inc)	89.94 +/- 1.45
Airmass coefficient 1 (a1)	1.48 +/- 0.51
Airmass coefficient 2 (a2)	-0.27 +/- 0.31
Scatter in the residuals of the lightcurve fit is	34.59 %
Best Comparison Star	#2 - [528, 271]
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.0933 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.048 vs 0.1301 (deviation 0.74σ)
Duration fit vs published	0.0933 d vs 0.0908 d (deviation 0.6σ)
Mid-time O–C	1.2 min
Depth SNR	3.4
Residual scatter	34.59 %

Light curve

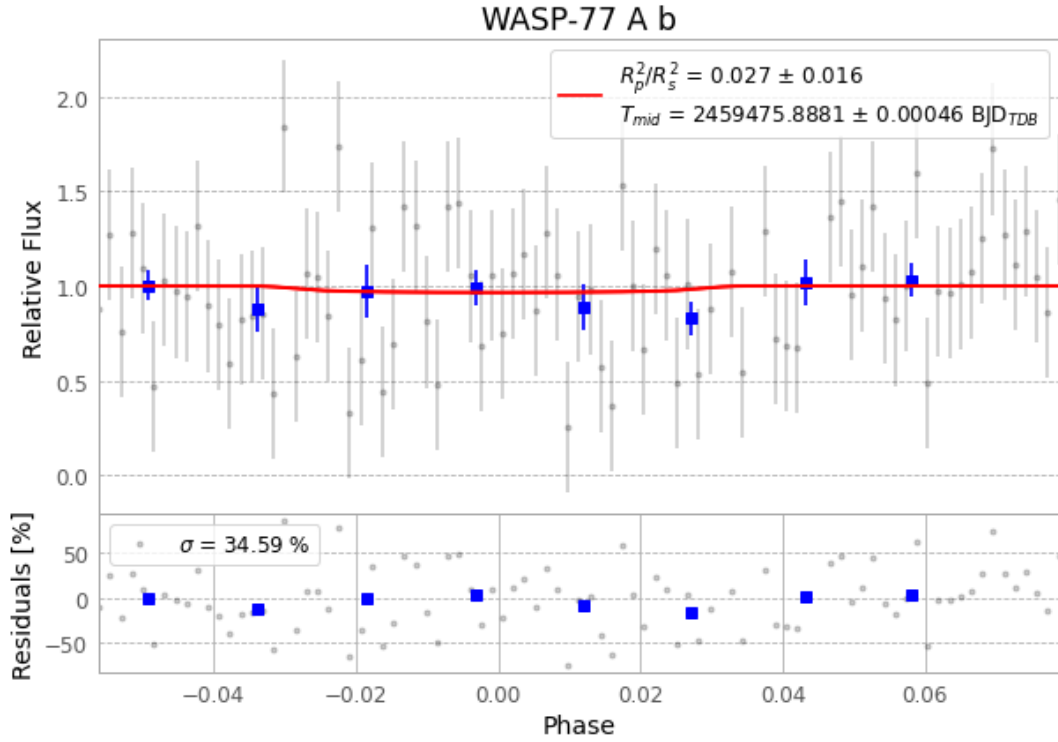


Figure 1 — FinalLightCurve_WASP-77 A b_2021-09-18.png (SHA-256 b88595f2ab951c98...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [387, 80], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9e4284f7ead34ec8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), WASP-77210918072112.FITS → WASP-77210918114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-210918.log (91b3f631111f...), FinalLightCurve_WASP-77 A b_2021-09-18.png (b88595f2ab95...), FinalLightCurve_WASP-77 A b_2021-09-18.csv (4aa5b3c32a9d...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 15:40Z.